

DAILY DRILLING REPORT SET

SYN-Assam-Duliajan-005 (W005) — SYN-Assam-Duliajan

1. REPORT SET IDENTIFICATION

Well ID	W005
Well Name	SYN-Assam-Duliajan-005
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude / Longitude	27.355745 N / 95.261269 E
Rig	SYN-RIG-E2000-07
Spud Date	28-Nov-2020
Final Report Date	09-Dec-2020
Daily Reports in Set	12 (Day 01 to Day 12)
Total Depth Reached	3,560 m
Primary Formation	Sandstone
Events Reported in Set	5

2. DEPTH PROGRESSION SUMMARY

Day	Date	From (m)	To (m)	Drilled (m)	Formation	Events
01	28-Nov-2020	0	535	535	Claystone	—
02	29-Nov-2020	535	845	310	Claystone / Shale	—
03	30-Nov-2020	845	1,265	420	Shale	Wellbore Instability
04	01-Dec-2020	1,265	1,645	380	Shale	—
05	02-Dec-2020	1,645	1,990	345	Shale	—
06	03-Dec-2020	1,990	2,200	210	Shale / Sandstone	Mud Loss, Wellbore Instability
07	04-Dec-2020	2,200	2,360	160	Sandstone	Cementing Issue, Mud Loss
08	05-Dec-2020	2,360	2,630	270	Sandstone	—
09	06-Dec-2020	2,630	2,885	255	Sandstone	—
10	07-Dec-2020	2,885	3,125	240	Sandstone	—
11	08-Dec-2020	3,125	3,350	225	Sandstone / Limestone	—
12	09-Dec-2020	3,350	3,560	210	Limestone	—

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). Depths, dates, parameters and events are generated from the prototype datasets and are not operational records of any company.

DAILY DRILLING REPORT — DAY 01

W005 · 28-Nov-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-11-28
Day Number: 1
Start Depth: 0 m
End Depth: 535 m
Drilled Interval: 535 m
Formation: Claystone
Mud Weight: 9.38 ppg
ROP: 32.1 m/hr
RPM: 142
WOB: 7.4 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Rig up completed and well spudded at 28-Nov-2020. Drilled 17-1/2" hole from surface to 535 m.
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

Geological observations.

Cuttings throughout the interval are claystone, consistent with the 0–782 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 9.38 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 535 m in Claystone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 02

W005 · 29-Nov-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-11-29
Day Number: 2
Start Depth: 535 m
End Depth: 845 m
Drilled Interval: 310 m
Formation: Claystone / Shale
Mud Weight: 10.03 ppg
ROP: 28.7 m/hr
RPM: 137
WOB: 10.3 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 310 m of section from 535 m to 845 m (16.5 hr on bottom).
- Deviation survey taken; hole angle within tolerance.
- Pulled out of hole, ran 13-3/8" casing to 780 m and cemented as programmed (Lead 12.6 ppg / Tail 15.8 ppg, cement to surface). Waited on cement and pressure tested.

Geological observations.

Formation change crossed during the tour: Claystone to 782 m, Shale to 2,008 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 10.03 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 845 m in Claystone. 13-3/8" casing cemented at 780 m; waiting on cement.

DAILY DRILLING REPORT — DAY 03						W005 · 30-Nov-2020
Well ID: W005 Well Name: SYN-Assam-Duliajan-005 Report Date: 2020-11-30 Day Number: 3 Start Depth: 845 m End Depth: 1265 m Drilled Interval: 420 m Formation: Shale Mud Weight: 10.60 ppg ROP: 17.4 m/hr RPM: 130 WOB: 15.0 klbf Events: Wellbore Instability at 916 m (High) NPT: 23.8 hours						
Operations performed. - Drilled 420 m of Shale from 845 m to 1,265 m (2.0 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Wellbore Instability handled at 916 m — 23.8 hr non-productive time. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 782–2,008 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 10.60 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0036	916	Shale	Wellbore Instability	High	23.8	Significant non-productive time; remedial operations and additional materials required.
End-of-day status. Depth 1,265 m in Shale. Drilling ahead. Non-productive time this tour 23.8 hr.						

DAILY DRILLING REPORT — DAY 04						W005 · 01-Dec-2020
Well ID: W005 Well Name: SYN-Assam-Duliajan-005 Report Date: 2020-12-01 Day Number: 4 Start Depth: 1265 m End Depth: 1645 m Drilled Interval: 380 m Formation: Shale Mud Weight: 11.17 ppg ROP: 16.2 m/hr RPM: 123 WOB: 17.9 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 380 m of Shale from 1,265 m to 1,645 m (22.5 hr on bottom). - Deviation survey taken; hole angle within tolerance.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 782–2,008 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.17 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day. None. Operations proceeded within the planned parameter envelope.						
End-of-day status. Depth 1,645 m in Shale. Drilling ahead. No non-productive time this tour.						
Prototype Training Record · Synthetic demonstration data						

DAILY DRILLING REPORT — DAY 05

W005 · 02-Dec-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-12-02
Day Number: 5
Start Depth: 1645 m
End Depth: 1990 m
Drilled Interval: 345 m
Formation: Shale
Mud Weight: 11.69 ppg
ROP: 15.0 m/hr
RPM: 119
WOB: 20.8 klbf
Events: None
NPT: 0.0 hours

Operations performed.

• Drilled 345 m of Shale from 1,645 m to 1,990 m (22.5 hr on bottom).

• Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 782–2,008 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.69 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,990 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 06

W005 · 03-Dec-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-12-03
Day Number: 6
Start Depth: 1990 m
End Depth: 2200 m
Drilled Interval: 210 m
Formation: Shale / Sandstone
Mud Weight: 11.86 ppg
ROP: 16.6 m/hr
RPM: 116
WOB: 22.0 klbf
Events: Mud Loss at 2159 m (High); Wellbore Instability at 2191 m (Medium)
NPT: 22.7 hours

Operations performed.

• Drilled 210 m of section from 1,990 m to 2,200 m (2.0 hr on bottom).

• Made connections and circulated bottoms-up at intervals; shakers monitored.

• Pulled out of hole, ran 9-5/8" casing to 2,010 m and cemented as programmed (Lead 12.9 ppg / Tail 15.8 ppg, TOC 660 m). Waited on cement and pressure tested.

• Mud Loss handled at 2,159 m — 11.1 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

• Wellbore Instability handled at 2,191 m — 11.6 hr non-productive time. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Geological observations.

Formation change crossed during the tour: Shale to 2,008 m, Sandstone to 3,176 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 11.86 ppg after treatment.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0037	2,159	Sandstone	Mud Loss	High	11.1	Significant non-productive time; remedial operations and additional materials required

Prototype Naming Record - Synthetic demonstration data

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DAILY DRILLING REPORT — DAY 07						W005 · 04-Dec-2020
Well ID: W005 Well Name: SYN-Assam-Duliajan-005 Report Date: 2020-12-04 Day Number: 7 Start Depth: 2200 m End Depth: 2360 m Drilled Interval: 160 m Formation: Sandstone Mud Weight: 12.13 ppg ROP: 16.3 m/hr RPM: 115 WOB: 23.1 klbf Events: Cementing Issue at 2278 m (Critical); Mud Loss at 2296 m (Low) NPT: 45.4 hours						
Operations performed. - Drilled 160 m of Sandstone from 2,200 m to 2,360 m (2.0 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored. - Cementing Issue handled at 2,278 m — 42.5 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage. - Mud Loss handled at 2,296 m — 2.9 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,008–3,176 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.13 ppg after treatment.						
Equipment / operational observations. Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0039	2,278	Sandstone	Cementing Issue	Critical	42.5	Major non-productive time; section objectives affected and specialist intervention required.
E0035	2,296	Sandstone	Mud Loss	Low	2.9	Minor non-productive time; operations resumed after routine parameter adjustment.
End-of-day status. Depth 2,360 m in Sandstone. Drilling ahead. Non-productive time this tour 45.4 hr.						

DAILY DRILLING REPORT — DAY 08						W005 · 05-Dec-2020
Well ID: W005 Well Name: SYN-Assam-Duliajan-005 Report Date: 2020-12-05 Day Number: 8 Start Depth: 2360 m End Depth: 2630 m Drilled Interval: 270 m Formation: Sandstone Mud Weight: 12.44 ppg ROP: 16.1 m/hr RPM: 113 WOB: 25.1 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 270 m of Sandstone from 2,360 m to 2,630 m (22.5 hr on bottom). - Deviation survey taken; hole angle within tolerance.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,008–3,176 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 12.44 ppg, properties held within programme, no losses or gains recorded.						
Equipment / operational observations. Solids-control equipment cleaned: shaker screens changed. No downtime.						

DAILY DRILLING REPORT — DAY 09

W005 · 06-Dec-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-12-06
Day Number: 9
Start Depth: 2630 m
End Depth: 2885 m
Drilled Interval: 255 m
Formation: Sandstone
Mud Weight: 12.83 ppg
ROP: 14.7 m/hr
RPM: 109
WOB: 26.2 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 255 m of Sandstone from 2,630 m to 2,885 m (22.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,008–3,176 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.83 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,885 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 10

W005 · 07-Dec-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-12-07
Day Number: 10
Start Depth: 2885 m
End Depth: 3125 m
Drilled Interval: 240 m
Formation: Sandstone
Mud Weight: 13.15 ppg
ROP: 13.9 m/hr
RPM: 108
WOB: 28.9 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 240 m of Sandstone from 2,885 m to 3,125 m (22.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,008–3,176 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 13.15 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Solids-control equipment cleaned; shaker screens changed. No downtime.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,125 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 11

W005 · 08-Dec-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-12-08
Day Number: 11
Start Depth: 3125 m
End Depth: 3350 m
Drilled Interval: 225 m
Formation: Sandstone / Limestone
Mud Weight: 13.74 ppg
ROP: 9.4 m/hr
RPM: 97
WOB: 33.0 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 225 m of section from 3,125 m to 3,350 m (22.5 hr on bottom).
- Deviation survey taken; hole angle within tolerance.

Geological observations.

Formation change crossed during the tour: Sandstone to 3,176 m, Limestone to 3,560 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 13.74 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,350 m in Limestone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 12

W005 · 09-Dec-2020

Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Report Date: 2020-12-09
Day Number: 12
Start Depth: 3350 m
End Depth: 3560 m
Drilled Interval: 210 m
Formation: Limestone
Mud Weight: 14.14 ppg
ROP: 7.5 m/hr
RPM: 89
WOB: 36.8 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 210 m of Limestone from 3,350 m to 3,560 m (16.5 hr on bottom).
- Deviation survey taken; hole angle within tolerance.
- Pulled out of hole, ran 7" casing to 3,560 m and cemented as programmed (Tail 15.8 ppg, TOC 1,860 m). Waited on cement and pressure tested.
- Reached driller's total depth of 3,560 m. Circulated hole clean, logged and secured the well.

Geological observations.

Cuttings throughout the interval are limestone, consistent with the 3,176–3,560 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 14.14 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,560 m in Limestone. 7" casing cemented at 3,560 m; waiting on cement.